



Guidelines

Hip Fracture Management

Scope

Site	Service/Department/Unit	Disciplines
Sir Charles Gairdner Osborne Park Health Care Group (SCGOPHCG)	Organisation Wide	All Staff

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Introduction

The Sir Charles Gairdner Osborne Park Health Care Group (SCGOPHCG) Hip Fracture Management Guideline has been written in accordance to the quality statements outlined in the Australian Commission on Safety and Quality on Health Care Hip Fracture Care Clinical Care Standard and aims to deliver best evidence based clinical practice in the care of patients with hip fractures.

1.0 Pre-Operative Assessment and Team Roles

1.1 Multidisciplinary Care

All hip fracture patients are admitted under the joint care of Orthopaedic Surgery and Geriatric Medicine. The Geriatrician will assume governance for medical decisions and the Orthopaedic Surgeon will assume governance for those decisions relating to surgical management. At all times a collaborative approach will prevail to ensure the best outcome for each individual patient.

The aim is to:

- Optimise patients' peri operative clinical condition
- Ensure prompt surgical fixation (operate within 24 hours of admission)
- Deliver early rehabilitation

All hip fracture patients will be cared for by a multidisciplinary team:

- They will be reviewed by a member of the Ortho-Geriatric team within 48 hours of admission and regularly post-operatively
- They will have nursing care according to the Hip Fracture Clinical Pathway
- They will be reviewed by the ward pharmacist on the first working day post admission
- They will be reviewed by a physiotherapist within 24 hours postoperatively
- They will have access to occupational therapy, social work, dietitian and speech pathology as required.

1.2 Cognitive Assessment

All hip fracture patients presenting to SCGH have a cognitive assessment (AMT4) performed in ED as per the "Fractured Hip Clinical Pathway".

All hip fracture patients at SCGH should be commenced on the SCGH Peri-operative Cognitive Impairment Pathway to screen for cognitive impairment and minimise the risk of peri-operative delirium.

For further information refer to the [SCGOPHCG Cognitive Impairment Management Clinical Practice Guideline](#).

1.3 Frailty Assessment

All hip fracture patients at SCGH have a frailty assessment on admission using the Clinical Frailty Scale as per the ANZHFRC.

1.3 Nutrition Assessment

All SCGOPHCG patients are screened for malnutrition on admission as per the Nursing Admission Assessment.

Patients identified as being at high risk of malnutrition are referred to the Dietitian for nutrition assessment and individualised intervention (see the [Malnutrition Action Plan for Nursing Staff](#) for more information).

Protein and energy oral nutritional supplements may be commenced by the Orthogeriatric Team if deemed appropriate

1.4 Goals of Patient Care

All hip fracture patients should have a “Goals of Patient Care” discussion preoperatively. This should include discussion on resuscitation status and limitations of treatment. This is to be documented on the Goals of Patient Care Form as per [SCGOPHCG Goals of Patient Care Policy](#).

1.5 Reporting to the Coroner

All deaths resulting from a fall must be reported to the Coroner’s Court on 9425 2900 before a death certificate can be issued. Following discussion of the case with the coroner you will be instructed whether or not a death certificate can be written. *Refer to IC0008/07 - Coroners Act 1996.*

1.6 Pathology Ordering

In keeping with “Choosing Wisely” principles pathology and radiology tests are to be ordered in line with the SCGH Orthopaedics Investigations Ordering Guideline (see [appendix 1](#)).

1.7 Pressure Injury Care

Hip fracture patients are at very high risk of developing a pressure injury while in hospital.

The following measures are to be implemented to decrease this risk:

- Daily skin inspections
- Pressure injury prevention mattress
- Heel protectors (if unable to wear graduated compression stockings)
- Heel elevators
- Sacral Meplilex Border dressing
- Frequent repositioning, sitting out of bed for meals, mobilise to the toilet and shower (avoid use of bed pans and bed baths).

For further information refer to the [SCGOPHCG Pressure Injury Prevention and Management Clinical Practice Guideline](#).

2.0 Pain Management

All hip fracture patients presenting to SCGH ED should have a pain assessment and analgesia administered (if required) within 30 minutes of arrival.

Older patients are very susceptible to the side effects of narcotic analgesia (e.g. sedation, delirium, constipation etc.) and the following basic prescribing principles should apply:

2.1 Regional Analgesia

Femoral Nerve Block

- All hip fracture patients are to receive a femoral nerve block under ultrasound guidance in the Emergency Department prior to transfer to the ward.

Femoral Nerve Catheter

- This is the preferred method of pre-operative analgesia for patients who may have a delay to theatre and the femoral nerve block is no longer effective
- To arrange this contact the duty anaesthetist on extension 71242 in hours and the on call anaesthetic registrar on pager 4823 after hours (after hours there may be insufficient staff available to perform the procedure).

2.2 Oral Analgesia

- Start with the lowest dose possible (START LOW)

- Do not escalate dosage rapidly (GO SLOW)
- ALL patients should also be charted for regular paracetamol (1 gram TDS – QID)
- Avoid polypharmacy:
 - prescribe **only one type of opioid**
 - prescribe only one type of immediate release opioid
- Avoid Codeine (Panadeine Forte), Morphine, Buprenorphine, Methadone, Gabapentinoids, NSAIDs and Tramadol.
- Suitable ORAL ANALGESIC regimes for the elderly are:
 - Tapentadol Slow release (SR) 50mg BD plus Oxycodone 2.5 – 5 mg 2 hourly prn; **or**
 - Targin (oxycodone 5mg / naloxone 2.5mg) BD plus Oxycodone 2.5 – 5 mg 2 hourly prn
- For the very frail elderly consider:
 - Targin (oxycodone 2.5mg / naloxone 1.25mg) BD plus Oxycodone 2.5 mg 2 hourly prn
- Patients will need to be prescribed regular aperients with this regime as described below.

Opioid tolerant patients may require higher doses of analgesia peri-operatively. It is best to initially continue their usual analgesic regime and prescribe higher doses of breakthrough analgesia. The Acute Pain Service should be consulted for patients with complex analgesic requirements such as those with chronic pain on complex analgesic regimes.

ALL NEW PERI-OPERATIVE ANALGESIC PRESCRIPTIONS SHOULD HAVE A REVIEW AND CEASE DATE DOCUMENTED ON THE MEDICATION CHART (EG : “for 2 weeks and then cease”)

3.0 Bladder & Bowel Management

3.1 Bowel Management

All patients that are prescribed narcotic analgesia must have regular aperients prescribed.

The recommended prescribing regime is:

- Coloxyl with Senna 2 nocte
- Lactulose 20mls nocte
- Two suppositories (1 glycerin and 1 bisacodyl) 3rd daily prn if bowels not open

Ensure patients have their bowels open at least every 3rd day.

3.2 Bladder Management

Hip fracture patients are **NOT** to receive indwelling urinary catheters (IDCs) unless clinically indicated (e.g. bladder stretch injury, monitoring urinary output).

Pre and post operatively patients are to have 4 – 6 hourly bladder scans and intermittent catheterisation (IMCs) as required (*Refer to SCGH Bladder Management and Urinary Catheterisation Nursing Practice Guideline and OPH Bladder Management and Urinary Catheterisation*).

A bladder scan is to be performed in the operating theatre at the end of the procedure. If the volume is >500ml an IMC is to be performed in theatre.

4.0 Prophylactic Antibiotics

RMO to chart pre-operatively:

- nasal mupirocin 2 % ointment inside both nostrils twice each day for five days (equivalent to a total of 10 doses), AND
- chlorhexidine 2 % or 4 % body wash once daily for 5 days.
- Once all recommended applications have been completed, discard tube and any remaining contents.

For further information, refer to the [SCGOPHCH Mupirocin Nasal Ointment for Decolonisation Clinical Practice Guideline](#).

Anaesthetist to chart in OT:

- Cefazolin 2gm IV within the 60minutes before surgical incision

For patients with immediate hypersensitivity to penicillins:

- Vancomycin 15mg/kg IV within 120minutes before surgical incision
- Suggested intraoperative redosing intervals for antibiotics commonly used for surgical antibiotic prophylaxis (Table 2.15) on eTG ([here](#)). Do not give additional doses once the procedure is completed.

Re-dosing of antibiotics for extended surgical procedures is as follows:

- a. Cefazolin – intraoperative re-dosing is required after 4 hours
- b. Vancomycin – intraoperative re-dosing is required after 12 hours

NB: The re-dosing interval is measured from the initial pre-operative dose, rather than the beginning of the operation.

Hip fracture patients having fixation with a total hip replacement (THR) or hemiarthroplasty are to have a further 2 doses of cephazolin 2gm post operatively.

NB: Although a single preoperative dose of surgical antibiotic prophylaxis is expected to be sufficient to prevent postoperative infection following total knee arthroplasty, there is insufficient evidence to show that a single dose of prophylaxis is as effective as 24 hours of prophylaxis. Postoperative doses can be considered but prophylaxis (intravenous or oral) should not continue beyond 24 hours.

5.0 Perioperative blood pressure management

Hip fracture patients are at risk of post-operative hypotension. This is often multifactorial and related to anaesthesia, blood loss and SIRs (Systemic Inflammatory Response to proximal long bone fracture and its manipulation resulting in distributive shock). To minimise the risk of post-operative hypotension the following measures are to be implemented:

- Antihypertensive medications should be withheld perioperatively if systolic BP is less than 130mmHg. (with the exception of those prescribed for other indications eg: beta blockers for rate control or ischaemic heart disease). Please write on the medication chart “WITHHOLD IF SBP <130mmHg”
- Midodrine 20mg orally stat is to be prescribed preoperatively. This is to be administered on the ward just prior to transfer to theatre with instruction to “WITHHOLD if SBP > 160mmHg”
- Midodrine 10mg orally 4 hourly is to be prescribed for 48 hours postop with instructions to “WITHHOLD if SBP >130mmHg”.

6.0 Perioperative fluid management

Preoperative Management

Clinical outcomes for surgical patients are improved when perioperative oral clear fluids are continued until the patient is sent to theatre.

Hip fracture patients are to follow the “Sip Til Send” protocol and fast from food 6 hours pre op and may have 200mls / hour of clear fluids orally until theatre. IV fluids are routinely not required preoperatively. See the [Preoperative care Nurse Practice Guideline](#) for further information.

Older patients often have poor fluid intake and hence low urine output prior to admission.

Many elderly patients also have poor cardiac function making them very susceptible to developing acute pulmonary oedema if given aggressive intravenous fluid therapy peri-operatively.

Urine output of 15-20ml/hr may be acceptable in the frail elderly and maintenance fluids of 80-100mls/hr may be more appropriate.

7.0 Postoperative fluid management

When reviewing the ‘hypotensive’ or ‘low urine output’ patient postoperatively each case needs to be reviewed on an individual basis. If the patient usually has a low blood pressure, is clinically well or asymptomatic, is not dehydrated and does not have deteriorating renal function they are unlikely to require aggressive fluid resuscitation.

8.0 Oxygen therapy

Oxygen saturations should be measured on all patients with hip fractures both on admission and routinely on the ward. Low saturations on room air should trigger a search for any treatable pathology (infection, LVF, PE, atelectasis).

Oxygen should be considered as a drug and if required, should be prescribed.

Oxygen should be prescribed as per [the Thoracic Society of Australia and New Zealand Oxygen Guidelines for Acute Oxygen Use in Adults](#), found on the Thoracic Org website.

- **For patients at risk of hypercapnaeic respiratory failure O2 sats on RA of $\geq 88\%$ are often acceptable and oxygen therapy should be titrated to a target range of 88-92%.**
- **In an emergency situation, oxygen therapy should never be withheld from patients.**

9.0 Venous thromboembolism (VTE) prophylaxis

All patients must have the VTE Risk Assessment completed on the medication chart.

All patients are to have VTE prophylaxis in the form of:

Limited period of immobilisation

- Patients are to have surgery as soon as possible after presentation to hospital or sustaining a hip fracture while in hospital (unless clinically contraindicated)
- Patients are to be mobilised as soon as possible after surgery and at least once a day thereafter (assuming the patient’s clinical condition and goals of care allow)

Mechanical prophylaxis

- Patients are to have graduated compression stockings applied preoperatively and worn throughout their hospital stay (unless clinically contraindicated)

- Intermittent pneumatic compression (**IPC**) devices are contraindicated in hip fracture patients due to their prolonged period of immobilisation prior to presentation

Pharmacological prophylaxis

- LMWH (enoxaparin) should be prescribed and administered according to the “standard” VTE prophylaxis guidelines (unless clinically contraindicated)
- daily administration of enoxaparin should be timed for 2000h (8pm).
- Patients who are not operated on immediately should, unless contraindicated, have their usual 20:00 Enoxaparin the night before surgery.

For further information refer to *SCGH Venous Thrombo-Embolism (VTE) Prophylaxis Policy* and *OPH Venous Thromboembolism Prophylaxis*.

10.0 Post-operative mobilisation

Patients are to be operated on with the aim of allowing them to fully weight bear (without restriction) in the immediate postoperative period.

Patients on a morning theatre list are to be mobilised that afternoon (i.e. Day 0 post-op)

Patients on an afternoon theatre list are to be mobilised early the next morning (i.e. Day 1 post op).

11.0 Antiplatelet and Anticoagulants Management

- Clearly document indication for the antiplatelet therapy
- In general, surgery should **NOT BE DELAYED** in hip fracture patients taking antiplatelet therapy.

11.1 Aspirin Therapy

- Aspirin therapy should be maintained throughout admission unless a contraindication arises (e.g. GI bleeding).

11.2 Antiplatelet Medication (Aspirin / Clopidogrel / Ticagrelor / Prasugrel) for Indications Other Than Vascular Stents

- **DO NOT** cease any dual antiplatelet therapy
- Withhold further dose pre-operatively. If the patient has had a recent atherothrombotic event (esp < 1 month) the treating physician / surgeon should be contacted prior to any alteration of antiplatelet therapy. Discuss with haematology about “safe for theatre” times.
- If the patient is at high risk of an ischaemic event or there is a contraindication to recommencing their antiplatelet therapy post-op consider covering with aspirin therapy (if no contraindication exists)
- Recommence 24 - 48 hours post-op if no surgical contraindication exists.

11.3 Antiplatelet Medication (Aspirin / Clopidogrel / Ticagrelor / Prasugrel) In Patients with Vascular Stents

- **DO NOT** cease any dual antiplatelet therapy
- If possible document:
 - 1) date of stent insertion
 - 2) type of stent inserted
 - 3) anatomical location of stent
 - 4) any specific recommendations from the interventionist at time of procedure.
- If the patient has a history of stent insertion within the last 12 months (or any other concerns), the treating physician / surgeon should be contacted prior to any

alteration of antiplatelet therapy. Discuss with haematology about “safe for theatre” times.

11.4 Warfarin Reversal

- **WITHHOLD** further doses of warfarin
- Clearly document
 1. indication for warfarin
 2. usual dose of warfarin
 3. date / time of last dose
- Take INR
- Prescribe BRIDGING ANTICOAGULATION if indicated as below
- Speak to cardiologist /on call haematologist for specific advice if the patient has a mechanical heart valve.

If going straight to theatre give:

- Vit K 10 mg IV (UNLESS MECHANICAL HEART VALVE)
- Beriplex dosing is guided by the INR. See table 1
- Recheck the INR 30 minutes post administration of Beriplex.

If going to theatre next day give:

- Vit K 10 mg IV now (UNLESS MECHANICAL HEART VALVUE)
- Repeat INR in am Beriplex dosing is guided by the INR. See table 1.

	Initial INR			
Target INR	1.5-2.5	2.6-3.5	3.6-10.0	>10.0
0.9-1.3	30 IU/kg	35 IU/kg	50 IU/kg	50 IU/kg
1.4-2.0	15 IU/kg	25 IU/kg	30 IU/kg	40 IU/kg

Table 1: Suggested dose of 4-Factor PCC to reverse warfarin pre-operatively (taken from Robinson et al 2025)

Note:

- When Vitamin K is given IV it has an onset of action within 6-8 hours.
- Beriplex contains small amounts of heparin and its use to reverse warfarin should be carefully considered in patients with a history of heparin-induced thrombocytopenia. Discuss with haematology.
- Dosing of Beriplex is capped at 100kg. Therefore, the maximum dose is 5000IU.

11.5 Bridging anticoagulation

Consider bridging anticoagulation in patients with a high thrombotic risk with therapeutic dose LMWH while the INR is <2.0 for the following patients:

- **Atrial Fibrillation (AF) and**
 - CHADS2 score of 5 or 6
 - Rheumatic valvular disease
 - Recent CVA / TIA (< 3 months)

Patients with AF who have a CHADS2 score of ≤ 4 and who have not had a stroke or TIA in last three months should not receive bridging (Bridge trial).

- **VTE (DVT/ PE)**
 - Recent VTE < 3 months OR Recurrent VTE despite therapeutic anticoagulation:
 - Discuss with haematology.
 - Consider an IVC filter if the VTE occurred within 1 month. Please discuss with haematology.

- **Mechanical Heart Valve (MHV)**

- Bridging therapy is recommended for patients with MHVs except those with a Bicuspid aortic valve and no other risk factors (Keeling, *et al* 2011, BCSH guidelines).

11.5.1 Prescription of bridging anticoagulation

- Start treatment dose LMWH (1mg/kg BD dosing regimen if normal renal function) when the INR falls to < 1.5.
- Consider UFH bridging in patients with CrCl<30ml/min.
- In patients who are receiving pre-operative treatment dose LMWH the last dose should be at least 24 h before surgery.
- Patients who do not require bridging anticoagulation should be prescribed prophylactic dose LMWH. This is to commence when the INR< 2 and charted at 2000hrs. It should be commenced preoperatively unless contraindicated.

11.6 Postoperative Management

Postoperative bridging (i.e. treatment dose LMWH) is not restarted until 48 hours after surgery and the surgeons are happy to do so. Thromboprophylaxis should be given in the interim.

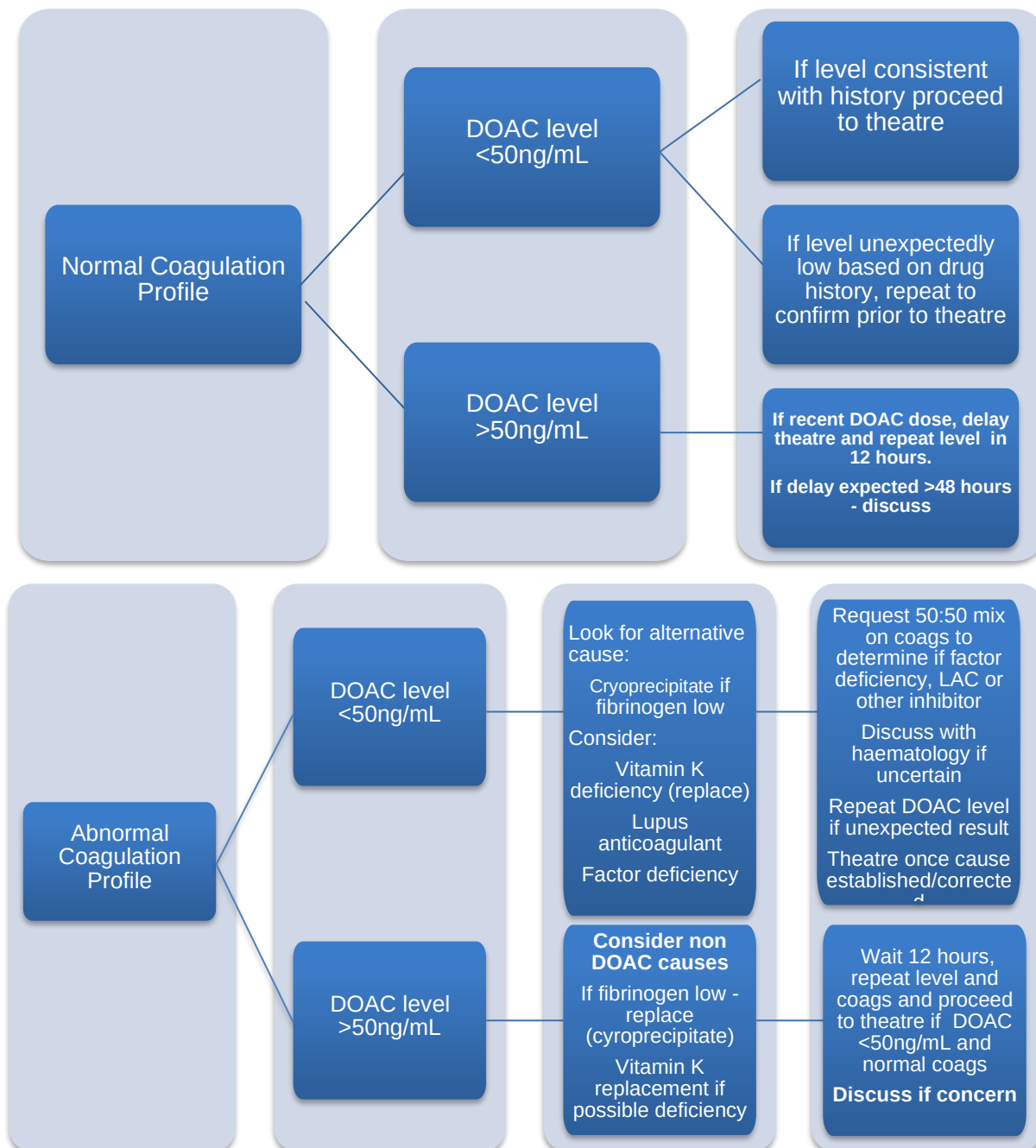
Recommencement of warfarin postoperatively may occur as soon as it is clinically safe to do so. Generally the patient's usual dose of warfarin should be recommenced. LMWH can be ceased when the INR is >2.0

11.7 Direct Oral Anticoagulant (DOAC) Management¹

- **WITHHOLD** further doses of DOAC
- Document;
 - DOAC name, dose, time of last dose
 - Indication for DOAC
- Request;
 - Coagulation profile
 - **Normal results** on coagulation tests do not exclude a therapeutic DOAC level.
 - DOAC drug level – request ASAP. This can be done on the coagulation sample and is available 24 hours.
 - U&E, LFT
- Determine safe time for surgery based on history, coagulation profile and drug level. (Refer to flow charts below)

NB If drug level is not in keeping with the clinical scenario (e.g.: time of last dose or renal function) review history and repeat drug level to confirm prior to theatre.

The half-life for DOACs is around 10 hours with normal renal function. Expected peak and trough DOAC levels vary between agents. Discuss with haematology if concern. If surgery is expected to be delayed >48 hours due to DOAC the risks and benefits of delay must be discussed. The on-call haematologist may be contacted for further advice.



11.8 Postoperative Resumption of DOAC

If bleeding is absent, resume DOAC 48-72 hours following hip fracture surgery. Prescribe prophylactic dose LMWH postoperatively while DOAC is being withheld.

For patients at high risk of thrombosis/thromboembolism an earlier reduced dose may be suitable (e.g. on the evening following surgery). Specialist haematologist advice is recommended.

11.9 When to Call Haematology

- Haemostatic products required, e.g. patient is bleeding or surgery is not deferrable.
- Coagulation results or drug levels are difficult to interpret.
- Significant renal dysfunction
- If surgery is expected to be delayed >48 hours due to DOAC clearance time
- If drug level is not in keeping with clinical scenario (eg : time of last dosage or renal function)
- Other significant haematological concerns

NB: A specific reversal agent Idarucizumab is available for Dabigatran. This is only for use in life threatening bleeding (e.g. multi trauma, intracerebral haemorrhage) or when urgent lifesaving surgery is required.

Refer to prescribing restrictions and treatment algorithm on the WA Statewide Medicine Formulary for idarucizumab [here](#).

12.0 Patient Blood Management

Patient Blood Management for hip fracture patients involves:

1. Iron Therapy
2. Tranexamic Acid Therapy
3. RBC Transfusion Therapy

12.1 Iron Therapy

Anaemia (Hb < 130g/L males, Hb < 120g/L females) is prevalent in the hip fracture population. Hip fracture surgery commonly results in a fall in haemoglobin level, triggering the need for allogenic blood transfusion. The frequency and volume of allogeneic blood transfusion following hip fracture surgery may be reduced by perioperative intravenous iron infusion.

Following consensus between orthopaedic surgery, anaesthetics and geriatric medicine, ***all hip fracture patients, will receive an intravenous infusion of 500 mg Iron Polymaltose (Ferrosig) intra-operatively unless a contraindication exists (see below)***

Contraindications to iron therapy:

- Iron overload syndromes (haemachromatosis, haemaglobinopathies)
- Patients requiring recurrent transfusions (e.g.: myelodysplasia)
- Polycythemia rubra vera

12.2 Tranexamic Acid (TXA) Therapy

There is moderate quality evidence that TXA reduces post-operative anaemia and the need for postoperative blood transfusion in hip fracture surgery, but its safety profile in this patient population is not well documented. Furthermore there is a paucity of data regarding the optimum therapeutic regime in this setting. Following a literature review and consensus between orthopaedic surgery, anaesthetics and geriatric medicine all hip fracture patients will receive **a single bolus of 15mg/kg IV TXA given at induction**. Patients **WILL NOT** receive subsequent doses.

Contraindications to TXA therapy:

- active intravascular clotting
- history of venous thromboembolic disease (DVT, PE)
- active coronary artery or cerebrovascular disease (event within the last 12 months)
- history of vascular stents
- history of retinal vein or artery occlusion
- congenital or acquired coagulation abnormality
- patients on anticoagulants
- active malignancy
- active genitourinary bleeding
- mechanical heart valves
- acquired defective colour vision
- subarachnoid haemorrhage
- patients with hypersensitivity to TXA

12.3 Transfusion Therapy

Red blood Cell (RBC) transfusion should NOT be dictated by haemoglobin alone but based on assessment of the patient's clinical status, comorbidities, intraoperative losses and risk of ongoing bleeding. Elderly patients with multiple co-morbidities tend not to tolerate anaemia well.

Preoperative:

- Group and Screen all patients

Hb < 90g/l – Notify Anaesthetics/Orthogeriatrics. (A significant preoperative anaemia should trigger a search for an underlying cause.)

Postoperative:

- Order FBP Day 1 post op (am round marked as **URGENT**)
- Transfuse the anaemic patient if:
 - they have symptoms that may be attributable to anaemia (hypoxia, shortness of breath, hypotension, hypovolemia, (pre)syncopal symptoms, tachycardia, delirium etc.)
 - they are actively bleeding
 - recent myocardial or cerebrovascular ischaemia (aim for Hb >100)
- Patients should only receive a Single Unit Transfusion followed by medical assessment +/- repeat formal lab Hb before consideration of a further unit
- Only order Day 2 post-op Hb if Day 1 Hb < 90 or clear clinical indication. For more information refer to *Appendix 1 of the SCGH Orthopaedic Investigation Ordering Guideline*.

13.0 Delirium Assessment and Management

Hip Fracture patients are at high risk of postoperative delirium.

Hip fracture patients are assessed regularly for delirium using the 4AT as per the “Nursing Admission Assessment”.

The aetiology is usually multifactorial. Although some factors are irreversible (pre-existing cognitive impairment, new environment and post-anaesthetic) potentially reversible causes should always be looked for and treated accordingly:

- Pain
- Constipation
- Urinary retention
- Dehydration / Poor nutritional intake
- Hypoxia
- Hypotension
- Electrolyte abnormality
- Occult sepsis (eg UTI, LRTI)
- Medication toxicity (anticholinergics, sedatives, opioids)
- Consider alcohol or benzodiazepine withdrawal – if this is suspected refer to AWS guidelines

Non-pharmacological measure should always be first line of management:

- Maintain low level sensory stimulation: soft sound and lighting
- Single room if possible
- Staff calmly to engage, distract and supervise the patient

- Avoid confrontation e.g. walk away, defer interventions or offer food and drink to diffuse tension
- Consider 1:1 companion
- Encourage family to stay and assist

If the patient poses a risk to themselves or others and non-pharmacological measures are not sufficient the following can be prescribed:

Risperidone 0.25 – 0.5 mg PO 4 hourly PRN (maximum 2 mg in 24 hours)

For patients with an extrapyramidal syndrome (eg: Parkinson's Disease) prescribe

Quetiapine 12.5 – 25 mg 4 hourly PO PRN (maximum 100 mg in 24 hours)

If very agitated and Risperidone/Quetiapine (as above) not adequate, add:

Olanzapine 2.5 – 5mg 4 hourly PO or IM (maximum 7.5mg in 24 hours)

- Always use oral before parental whenever possible.
- Avoid benzodiazepines as they may worsen delirium and increase falls risk
- **Remember to treat the underlying cause of the delirium**

14.0 Osteoporosis Assessment and Management

All patients admitted with a minimal trauma fracture should be investigated for underlying osteoporosis.

Serum Calcium and Vitamin D levels should be ordered on all minimal trauma fracture patients on admission.

Ensure the patient has adequate dietary calcium intake (1000-1300 mg per day).

If inadequate consider calcium supplementation with 600mg daily to bd

Ensure the patient has a Vitamin D level > 50 nmol/L.

If Vitamin D deficient prescribe supplementation as detailed below:

- Mild deficiency (30-49 nmol/L) : cholecalciferol 1000 - 2000 IU/day
- Moderate to severe deficiency (<30 nmol/L) : 3000-5000 IU/day for 6-12 weeks, followed by maintenance dose of 1,000-2,000 IU per day

Further osteoporosis treatment will be initiated in consultation with the Orthogeriatric team.

Patients identified as requiring outpatient follow-up regarding their bone health need an eReferral to Fragile Bone Clinic.

15.0 Falls Risk Assessment

All patients admitted with a fall and fracture should have a falls assessment as per the [NMHS Falls Risk Management Policy](#) and [SCGOPHCG Falls Management Clinical Practice Guidelines](#).

This should include an assessment and all appropriate interventions to mitigate identified risks regarding:

- Circumstances of the presenting fall
- Falls history
- Medication review to minimise polypharmacy and eliminate high risk medications
- Problems with elimination
- Cognitive Impairment (as per 12.0)
- Known vision impairment
- Heart rate and rhythm (History and ECG review)

- Postural hypotension (History and BP review)
- Patients identified by the Orthogeriatric team as requiring outpatient follow-up regarding their falls risk need an eReferral to Falls Clinic.
- For more information, refer to [WA DoH Website for Falls Specialist Service – information for providers and referrers.](#)

Compliance, monitoring and evaluation

Evaluation, audit and feedback processes are required to monitor compliance with this guideline as well as the Australian Commission Clinical Care Standards.

KPI performance target data is collected and monitored through the WA Hip Fracture data base (QoCR) and ANZ Hip Fracture Registry.

KPI Performance Targets include:

- Percentage of patients with pain assessment within 30 minutes of ED presentation
- Average time in Emergency Department
- Percentage of patients with Pre-operative medical assessment
- Percentage of patients with Pre-operative cognitive assessment
- Percentage of patients to theatre within 48 hours of presentation
- Percentage of patients receiving femoral nerve block
- Percentage of patients seen by a physiotherapist day one post operatively
- Average length of stay in hospital

Related Documents/Resources

- [WA DoH Website for Falls Prevention Health Network](#)
- [Thoracic Society of Australia and New Zealand Oxygen Guidelines for Acute Oxygen Use in Adults](#)
- [Australian Commission on Safety and Quality in Health Care Hip Fracture Care Clinical Care Standard](#)
- [Australian and New Zealand Guideline for Hip Fracture Care on ANZHFC website](#)
- [NMHS Falls Risk Management Policy](#)
- [SCGOPHCG Falls Management Clinical Practice Guidelines.](#)
- [SCGOPHCG Medicines Management Policy](#)
- [SCGOPHCG Pressure Injury Prevention and Management Clinical Practice Guideline.](#)
- [SCGOPHCG Mupirocin Nasal Ointment for Decolonisation Clinical Practice Guideline](#)
- [SCGOPHCG Cognitive Impairment Management Clinical Practice Guideline.](#)
- [SCGOPHCG Goals of Patient Care Policy.](#)
- [SCGOPHCG Malnutrition Action Plan for Nursing Staff](#)
- [WA Statewide Medicines Formulary Policy](#)
- [Discharge Analgesia Prescribing Guideline](#)
- [Post Fall Guidelines for Western Australian Health Care Settings](#)

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Std 3: Preventing and Controlling Healthcare Associated Infection



Std 4: Medication Safety



Std 5: Comprehensive Care



Std 8: Recognising and Responding to Acute Deterioration



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Appendix 1: SCGH Orthopaedics Investigations Ordering Guideline

SCGH Orthopaedics Investigation Ordering Guidelines

Choosing Wisely – for hip fracture and other orthogeriatric patients

PRE-OP

Routine	1. FBC	
	2. UEC	
	3. G+H (only for hip fracture patients going to theatre)	
Others	Coags	Only if specific clinical indication (e.g. severe liver disease, known coagulopathy, on DOAC*) *If on DOAC also order DOAC level
	LFTs	Only if specific clinical indication (e.g. liver disease or on hepatotoxic meds)
	INR	For those on warfarin – do not need full coagulation profile!
	CXR	Not routine – only if clinical indication (known respiratory disease or respiratory signs/symptoms)

POST-OP

Day 1	FBC	
	UEC	
	XR	Only if requested by surgeon on op report
	Ca + Vit D	For minimal trauma # patients <u>ONLY</u> (please order only <u>ONCE</u> during admission)
Day 2	FBC	Only if day 1 post-op Hb < 90 or if clinical indication (e.g. blood loss, hypotension, recent cerebral/cardiac ischaemia)
	UEC	Only if monitoring an abnormality (e.g. AKI, electrolyte abnormality)
> Day 3	Only if clinical indication – NO routine bloods needed if everything is stable day 2 post-op	

OTHER

MSU	To exclude UTI – only if abnormal UA (e.g. protein, leukocytes or nitrites)
CRP	<u>NOT</u> routine post-op, only order for suspected infection Once established on antibiotic treatment, twice weekly CRP is adequate for monitoring response to therapy
TSH	Order <u>ONLY</u> TSH for investigating thyroid dysfunction Do not order full TFT panel first line!